WSABI Summer Research Lab	**Summer 2026**

Lecture 1: Simple Linear Regression

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1.1 Motivating Question: Prediction and Uncertainty

Suppose we have access to each MLB player's 2020 and 2021 batting averages from the Lahman Database, restricting to players with at least 50 at-bats in each season, and no other information. How would we predict a player's 2021 batting average from their 2020 one? How uncertain should we be about that prediction?

This lecture has two main goals:

1. build a line that gives a good **point prediction**;
2. quantify **uncertainty** around both the fitted line and the prediction for an individual player.

Generally, a good idea is to start with exploratory data analysis. Let's plot the data and see what we learn.

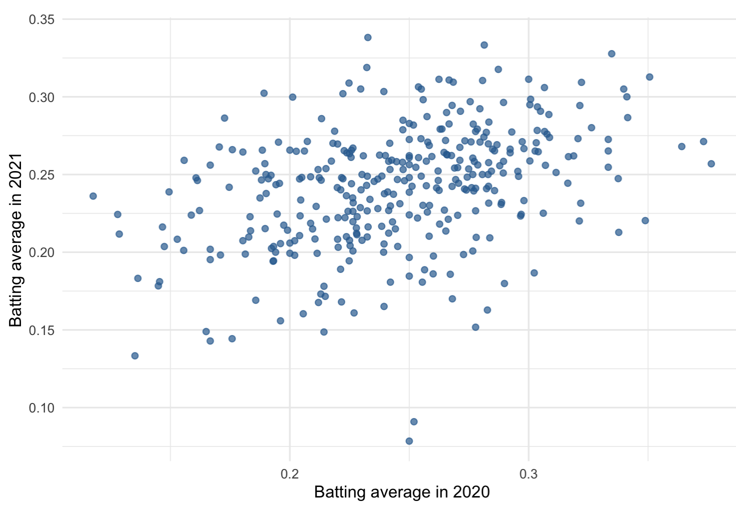

Figure 1.1: Each dot is one MLB player, with 2020 batting average on the horizontal axis and 2021 batting average on the vertical axis.

What does the relationship look like?

- Somewhat linear with a positive slope.
- Positive relationship: a higher 2020 BA is associated with a higher 2021 BA.
- Noisy relationship: 2020 BA does not perfectly determine 2021 BA.
- You might imagine drawing a best fit line through the points.

Figure 1.2: Same player-level dots as Figure 1.1, with the red line showing the fitted least-squares regression line.

Let's plot that line of best fit.

This relationship is not perfect. There is correlation, but there is also a lot of noise in the data. Our first question is: how did we get this line? Our second question is: after we get the line, how much uncertainty should we attach to its predictions?

1.2 Setting Up the Model

Index each baseball player by $i = 1, \ldots, n$. Let $X_i = BA_i^{(2020)}$ be our independent variable, also called the predictor, and let $Y_i = BA_i^{(2021)}$ be our dependent variable, also called the response. Assuming a **linear conditional mean** between each X_i and its corresponding Y_i, we can write

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,$$

where β_0 is an unknown constant intercept, β_1 is an unknown constant slope, and ϵ_i is a random error term.

A common simple linear regression model treats the observed predictor values as fixed and assumes

$$\mathbb{E}[\epsilon_i \mid X_1, \ldots, X_n] = 0, \qquad \mathrm{Var}(\epsilon_i \mid X_1, \ldots, X_n) = \sigma^2,$$

with independent errors across players. The conditional mean-zero assumption is what lets us say

$$\mathbb{E}[Y_i \mid X_i] = \beta_0 + \beta_1 X_i.$$

This is the "true" underlying regression line, but we do not know the values of β_0 and β_1. We therefore estimate them from the data.

It is useful to distinguish **errors** from **residuals**:

$$\epsilon_i = Y_i - \mathbb{E}[Y_i \mid X_i] \qquad \text{is the unobserved model error,}$$

while

$$e_i = Y_i - \widehat{Y}_i \qquad \text{is the observed residual after fitting the model.}$$

We never observe the true errors ϵ_i because we do not know the true line. We do observe the residuals e_i because we can compute the fitted line.

1.3 Estimating Model Parameters

Definition 1.1 (Ordinary Least Squares). *Ordinary least squares chooses $\widehat{\beta}_0$ and $\widehat{\beta}_1$ to minimize the* ***residual sum of squares (RSS)****.*

To avoid confusing candidate values with the true parameters, write b_0 and b_1 for possible intercept and slope values. The residual sum of squares for these candidate values is

$$\mathrm{RSS}(b_0, b_1) = \sum_{i=1}^{n} \left(Y_i - (b_0 + b_1 X_i)\right)^2 .$$

The least-squares estimates are the values that minimize this objective:

$$\begin{aligned}\widehat{\beta}_0, \widehat{\beta}_1 &= \underset{(b_0,b_1)}{\arg\min}\ \mathrm{RSS}(b_0, b_1) \\ &= \underset{(b_0,b_1)}{\arg\min}\ \sum_{i=1}^{n} \left(Y_i - (b_0 + b_1 X_i)\right)^2 .\end{aligned}$$

Visually, we can think of the RSS as the sum of the squared vertical distances between the observed points and the fitted line. In Figure 1.3, these vertical distances are the residuals.

Figure 1.3: Blue dots are observed players, the red line is the fitted regression line, and orange vertical segments show residuals (observed minus fitted values).

We can solve the least-squares problem with calculus by setting the partial derivatives with respect to b_0 and b_1 equal to zero. First,

$$\begin{aligned}\frac{\partial}{\partial b_0}\mathrm{RSS}(b_0, b_1) &= \sum_{i=1}^{n} -2\left(Y_i - (b_0 + b_1 X_i)\right) = 0 \\ &\implies \frac{1}{n}\sum_{i=1}^{n} b_0 = \frac{1}{n}\sum_{i=1}^{n}(Y_i - b_1 X_i) \\ &\implies b_0 = \bar{Y} - b_1 \bar{X}.\end{aligned}$$

Evaluating at the minimizer $(b_0, b_1) = (\widehat{\beta}_0, \widehat{\beta}_1)$ gives

$$\widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{X}. \tag{1.1}$$

Here $\bar{X} = n^{-1} \sum_{i=1}^n X_i$ and $\bar{Y} = n^{-1} \sum_{i=1}^n Y_i$ are the sample means of the predictor and response variables, respectively.

Second,

$$\begin{aligned}
\frac{\partial}{\partial b_1} \operatorname{RSS}(b_0, b_1) &= \sum_{i=1}^n -2X_i \left(Y_i - (b_0 + b_1 X_i)\right) = 0 \\
&\implies \frac{1}{n} \sum_{i=1}^n X_i Y_i - b_0 \frac{1}{n} \sum_{i=1}^n X_i - b_1 \frac{1}{n} \sum_{i=1}^n X_i^2 = 0 \\
&\implies \frac{1}{n} \sum_{i=1}^n X_i Y_i - b_0 \bar{X} - b_1 \frac{1}{n} \sum_{i=1}^n X_i^2 = 0.
\end{aligned}$$

Now evaluate the first-order condition at the least-squares estimates $b_0 = \widehat{\beta}_0$ and $b_1 = \widehat{\beta}_1$:

$$0 = \frac{1}{n} \sum_{i=1}^n X_i Y_i - \widehat{\beta}_0 \bar{X} - \widehat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2.$$

Substituting $\widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{X}$ from Equation 1.1, we get

$$\begin{aligned}
0 &= \frac{1}{n} \sum_{i=1}^n X_i Y_i - (\bar{Y} - \widehat{\beta}_1 \bar{X})\bar{X} - \widehat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2 \\
&= \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y} + \widehat{\beta}_1 \bar{X}^2 - \widehat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2.
\end{aligned}$$

Now collect the terms involving $\widehat{\beta}_1$ on one side and the remaining terms on the other side:

$$\widehat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \right) = \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}.$$

Assuming the X_i's are not all identical, the quantity in parentheses is positive, so we can divide by it:

$$\widehat{\beta}_1 = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}.$$

This expression is easier to interpret after rewriting both the numerator and denominator in centered form. First, because $\bar{Y}$ and $\bar{X}$ are constants,

$$\begin{aligned}
\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y} &= \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y} - \bar{X}\bar{Y} + \bar{X}\bar{Y} \\
&= \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{Y} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) - \bar{X} \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) + \bar{X}\bar{Y} \\
&= \frac{1}{n} \sum_{i=1}^n \left(X_i Y_i - X_i \bar{Y} - \bar{X} Y_i + \bar{X}\bar{Y} \right) \\
&= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}).
\end{aligned}$$

Similarly,

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2 &= \frac{1}{n}\sum_{i=1}^{n} X_i^2 - 2\bar{X}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) + \bar{X}^2 \\ &= \frac{1}{n}\sum_{i=1}^{n}\left(X_i^2 - 2\bar{X}X_i + \bar{X}^2\right) \\ &= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.\end{aligned}$$

Therefore, the slope can be written as

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}. \tag{1.2}$$

So we have closed-form expressions for $\widehat{\beta}_0$ and $\widehat{\beta}_1$. But what do these coefficients mean?

1.4 Interpreting Model Coefficients

1.4.1 Covariance

We first define covariance.

Definition 1.2 (Covariance)**.** *The covariance between two random variables X and Y is*

$$\begin{aligned}\sigma_{XY} = \mathrm{Cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

Note that if $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, then $\sigma_{XY} = \mathbb{E}[XY]$.

- Positive covariance: when X is above average, Y tends to be above average, and when X is below average, Y tends to be below average.
- Negative covariance: when X is above average, Y tends to be below average, and when X is below average, Y tends to be above average.

We proceed with some theorems that you will prove in your homework.

Theorem 1.3. *If X and Y are independent, then $\sigma_{XY} = 0$.*

Theorem 1.4. *Suppose $(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. draws from the joint distribution of (X, Y). The* ***sample covariance*** *between X and Y, defined as*

$$S_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}),$$

is an unbiased estimator of σ_{XY}.

Theorem 1.5. *Suppose $X_1, \ldots, X_n$ are i.i.d. draws from the distribution of X. The* ***sample variance*** *of X, defined as*

$$S_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

is an unbiased estimator of $\sigma_X^2 = \mathrm{Var}(X)$.

We are now ready to define the correlation between two random variables.

1.4.2 Correlation

Definition 1.6 (Correlation)**.** *The correlation between two random variables X and Y is*

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

It is a normalized version of the covariance, and is always between -1 and 1.

Definition 1.7 (Sample Correlation)**.** *The sample correlation between X and Y is*

$$r_{XY} = \frac{S_{XY}}{S_X S_Y} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}.$$

We can use the sample correlation to measure the strength of the linear relationship between X and Y. Consider our estimate $\widehat{\beta}_1$ from Equation 1.2:

$$\begin{aligned}
\widehat{\beta}_1 &= \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} \\
&= \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}} \cdot \frac{\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}} \\
&= r_{XY} \cdot \frac{S_Y}{S_X}.
\end{aligned}$$

This is a key interpretation: **the slope is correlation translated into the units of Y per unit of X**. It is also the sample version of the population formula derived in Appendix A.

We can also reciprocally express the sample correlation in terms of $\widehat{\beta}_1$:

$$r_{XY} = \widehat{\beta}_1 \frac{S_X}{S_Y}.$$

So if X and Y have the same scale, meaning the same sample standard deviations, then the sample correlation is simply the slope of the linear regression line.

1.4.3 Standardizing Variables

Sometimes it is useful to standardize variables before fitting a regression. Standardizing means subtracting the sample mean and dividing by the sample standard deviation.

Standardizing Only the Predictor

First suppose we standardize only the predictor:

$$X_i^* = \frac{X_i - \bar{X}}{S_X}.$$

The standardized predictor X_i^* has sample mean 0 and sample standard deviation 1. A one-unit increase in X_i^* therefore means a one-standard-deviation increase in the original predictor X_i.

Now regress Y_i on X_i^*:

$$Y_i = \gamma_0 + \gamma_1 X_i^* + e_i.$$

This is the same linear relationship as before, but the predictor is measured in standard-deviation units rather than original units.

The least-squares slope is

$$\widehat{\gamma}_1 = \frac{\sum_{i=1}^n (X_i^* - \bar{X}^*)(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i^* - \bar{X}^*)^2}.$$

Since $\bar{X}^* = 0$, this becomes

$$\begin{aligned}
\widehat{\gamma}_1 &= \frac{\sum_{i=1}^n X_i^*(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i^*)^2} \\
&= \frac{\sum_{i=1}^n \frac{X_i - \bar{X}}{S_X}(Y_i - \bar{Y})}{\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{S_X}\right)^2} \\
&= \frac{\frac{1}{S_X}\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\frac{1}{S_X^2}\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= S_X \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= S_X \widehat{\beta}_1.
\end{aligned}$$

Using $\widehat{\beta}_1 = r_{XY} S_Y / S_X$, we also have

$$\widehat{\gamma}_1 = r_{XY} S_Y.$$

So when only X is standardized, the slope still has units of Y:

A one-standard-deviation increase in X is associated with an estimated $r_{XY} S_Y$ increase in Y.

For example, if X is 2020 batting average and Y is 2021 batting average, then this slope tells us the predicted change in 2021 batting average associated with a one-standard-deviation increase in 2020 batting average.

Standardizing Both the Predictor and the Response

If we standardize both variables,

$$X_i^* = \frac{X_i - \bar{X}}{S_X}, \qquad Y_i^* = \frac{Y_i - \bar{Y}}{S_Y},$$

then both variables have sample mean 0 and sample standard deviation 1.

If we regress Y_i^* on X_i^*,

$$Y_i^* = \theta_0 + \theta_1 X_i^* + e_i^*,$$

then the fitted slope is

$$\widehat{\theta}_1 = r_{XY}.$$

Thus, correlation is the slope of the regression line when both variables have been standardized.

The interpretation is:

A one-standard-deviation increase in X is associated with an estimated r_{XY} standard-deviation increase in Y.

Standardizing changes the units of the coefficients, but it does not change the underlying linear relationship. Standardizing only X is useful when we want to keep Y in its original units. Standardizing both X and Y is useful when we want a unit-free measure of the strength of the linear relationship.

1.5 Back to Our Batting Average Model

In Section 1.2, we defined our model as

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,$$

where Y_i is the dependent variable, X_i is the independent variable, β_0 is the intercept, β_1 is the slope, and ϵ_i is the error term. Letting $X = BA^{(2020)}$ and $Y = BA^{(2021)}$, we now know that $\widehat{\beta}_1$ measures how strongly $BA^{(2020)}$ and $BA^{(2021)}$ move together, translated into the scale of batting average.

We use `R` to fit our linear regression model.

```
# fit simple linear regression
model = lm(BA_2021 ~ BA_2020, data = ba_data)

# display coefficient estimates and uncertainty summaries
summary(model)
confint(model)
```

From this output, we can see that $\widehat{\beta}_1 = 0.25$. This means that a batting average increase of 0.020 in 2020 is associated with a predicted batting average increase of 0.005 in 2021.

1.6 Regression to the Mean

If $X_i > \bar{X}$, then $X_i = \bar{X} + \delta$, with $\delta > 0$. Then it follows that

$$\begin{aligned}
\widehat{Y}_i &= \widehat{\beta}_0 + \widehat{\beta}_1 X_i \\
&= (\bar{Y} - \widehat{\beta}_1 \bar{X}) + \widehat{\beta}_1 X_i \\
&= \bar{Y} + \widehat{\beta}_1 (X_i - \bar{X}) \\
&= \bar{Y} + \widehat{\beta}_1 \delta.
\end{aligned}$$

In our example, $\widehat{\beta}_1 = 0.25$, therefore

$$\widehat{Y}_i = \bar{Y} + \frac{\delta}{4}.$$

This is an example of **regression to the mean**. $BA_i^{(2020)}$ is δ greater than $\bar{BA}^{(2020)}$, but $\widehat{BA}_i^{(2021)}$ is only $\frac{\delta}{4}$ greater than $\bar{BA}^{(2021)}$. In other words, if a player's 2020 batting average is δ above the 2020 sample mean, then our predicted 2021 batting average is only $\widehat{\beta}_1\delta$ above the 2021 sample mean. Since $\widehat{\beta}_1 = 0.25$, the prediction is pulled strongly back toward the 2021 mean.

This is a statement about the **point prediction**. It does not say the player's 2021 batting average must be close to the line. Individual players can still have large positive or negative residuals. That motivates our next question: how uncertain is the fitted line, and how uncertain is a prediction for one player?

1.7 Uncertainty in Regression

The fitted value $\widehat{Y}_i$ is an estimate, not a guarantee. In simple linear regression there are three useful layers of uncertainty:

1. typical noise around the fitted line;
2. uncertainty in the estimated slope;
3. uncertainty in a new prediction.

For this section, we treat the observed predictor values $X_1, \dots, X_n$ as fixed and assume

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \qquad \mathbb{E}[\epsilon_i \mid X_1, \dots, X_n] = 0, \qquad \mathrm{Var}(\epsilon_i \mid X_1, \dots, X_n) = \sigma^2,$$

with $\epsilon_1, \dots, \epsilon_n$ conditionally independent. If the errors are normally distributed, then the confidence intervals below are exact. Without normality, they are often still useful as large-sample approximations.

1.7.1 Residual Standard Error: Typical Noise Around the Line

The model says that individual observations vary around the true line:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i.$$

The parameter σ measures the typical size of the error term ϵ_i. Since we do not observe the true errors, we estimate them using residuals:

$$e_i = Y_i - \widehat{Y}_i = Y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 X_i).$$

The residual sum of squares is

$$\mathrm{RSS}(\widehat{\beta}_0, \widehat{\beta}_1) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y_i - \widehat{\beta}_0 - \widehat{\beta}_1 X_i\right)^2.$$

A natural estimate of σ^2 is the average squared residual. However, we do not divide by n. We divide by $n-2$ because we used the data to estimate two parameters, the intercept and the slope. Thus,

$$\widehat{\sigma}^2 = \frac{\mathrm{RSS}(\widehat{\beta}_0, \widehat{\beta}_1)}{n-2},$$

and the residual standard error is

$$\widehat{\sigma} = \sqrt{\frac{\mathrm{RSS}(\widehat{\beta}_0, \widehat{\beta}_1)}{n-2}}.$$

Interpretation: $\widehat{\sigma}$ estimates the typical vertical distance between an observed point and the fitted line. A small $\widehat{\sigma}$ means the points are tightly clustered around the line. A large $\widehat{\sigma}$ means individual outcomes are noisy even if the trend is real.

1.7.2 Uncertainty in the Slope

Let

$$S_{xx} = \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

From Equation 1.2, the estimated slope is

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})}{S_{xx}}.$$

We want to understand how much $\widehat{\beta}_1$ varies from sample to sample.

Using the model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, we can rewrite $\widehat{\beta}_1$ in terms of the true slope and the errors. First, note that

$$\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{\epsilon}, \qquad \text{where} \qquad \bar{\epsilon} = \frac{1}{n} \sum_{i=1}^{n} \epsilon_i.$$

Therefore,

$$\begin{aligned} Y_i - \bar{Y} &= (\beta_0 + \beta_1 X_i + \epsilon_i) - (\beta_0 + \beta_1 \bar{X} + \bar{\epsilon}) \\ &= \beta_1 (X_i - \bar{X}) + (\epsilon_i - \bar{\epsilon}). \end{aligned}$$

Substituting this into the formula for $\widehat{\beta}_1$ gives

$$\begin{aligned} \widehat{\beta}_1 &= \frac{\sum_{i=1}^{n} (X_i - \bar{X}) \left[\beta_1 (X_i - \bar{X}) + (\epsilon_i - \bar{\epsilon}) \right]}{S_{xx}} \\ &= \frac{\beta_1 \sum_{i=1}^{n} (X_i - \bar{X})^2 + \sum_{i=1}^{n} (X_i - \bar{X})(\epsilon_i - \bar{\epsilon})}{S_{xx}}. \end{aligned}$$

The first term simplifies because $\sum_{i=1}^{n} (X_i - \bar{X})^2 = S_{xx}$. For the second term,

$$\begin{aligned} \sum_{i=1}^{n} (X_i - \bar{X})(\epsilon_i - \bar{\epsilon}) &= \sum_{i=1}^{n} (X_i - \bar{X}) \epsilon_i - \bar{\epsilon} \sum_{i=1}^{n} (X_i - \bar{X}) \\ &= \sum_{i=1}^{n} (X_i - \bar{X}) \epsilon_i, \end{aligned}$$

because $\sum_{i=1}^{n} (X_i - \bar{X}) = 0$. Therefore,

$$\widehat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n} (X_i - \bar{X}) \epsilon_i}{S_{xx}}. \tag{1.3}$$

Equation 1.3 shows that $\widehat{\beta}_1$ equals the true slope plus a weighted average of the errors. Taking conditional expectation of Equation 1.3 given $X_1, \dots, X_n$, each term $(X_i - \bar{X})$ is fixed and $\mathbb{E}[\epsilon_i \mid X_1, \dots, X_n] = 0$. Therefore,

$$\mathbb{E}[\widehat{\beta}_1 \mid X_1, \dots, X_n] = \beta_1.$$

So $\widehat{\beta}_1$ is unbiased for β_1 under this model.

Now compute its variance. Since the X_i's are being treated as fixed and the errors are independent,

$$\begin{aligned}
\mathrm{Var}(\widehat{\beta}_1 \mid X_1, \dots, X_n) &= \mathrm{Var}\left(\frac{\sum_{i=1}^n (X_i - \bar{X})\epsilon_i}{S_{xx}} \,\middle|\, X_1, \dots, X_n \right) \\
&= \frac{1}{S_{xx}^2} \mathrm{Var}\left(\sum_{i=1}^n (X_i - \bar{X})\epsilon_i \,\middle|\, X_1, \dots, X_n \right) \\
&= \frac{1}{S_{xx}^2} \sum_{i=1}^n (X_i - \bar{X})^2 \, \mathrm{Var}(\epsilon_i \mid X_1, \dots, X_n) \\
&= \frac{1}{S_{xx}^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sigma^2 \\
&= \frac{\sigma^2 S_{xx}}{S_{xx}^2} \\
&= \frac{\sigma^2}{S_{xx}}. \qquad (1.4)
\end{aligned}$$

Since σ is unknown, we replace it with $\widehat{\sigma}$. Thus, the standard error of the estimated slope is

$$SE(\widehat{\beta}_1) = \sqrt{\frac{\widehat{\sigma}^2}{S_{xx}}} = \frac{\widehat{\sigma}}{\sqrt{S_{xx}}} = \frac{\widehat{\sigma}}{S_X \sqrt{n-1}}.$$

A 95% confidence interval for the slope is

$$\widehat{\beta}_1 \pm t_{n-2,0.975} SE(\widehat{\beta}_1).$$

Interpretation: $\widehat{\beta}_1 = 0.25$ is our best estimate of the slope, but the confidence interval tells us which slope values are reasonably compatible with the observed data.

1.7.3 Uncertainty for a New Prediction

Now suppose a new player has 2020 batting average x_0. The point prediction is

$$\widehat{y}_0 = \widehat{\beta}_0 + \widehat{\beta}_1 x_0.$$

Using $\widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{X}$, we can rewrite this as

$$\begin{aligned}
\widehat{y}_0 &= \bar{Y} - \widehat{\beta}_1 \bar{X} + \widehat{\beta}_1 x_0 \\
&= \bar{Y} + \widehat{\beta}_1 (x_0 - \bar{X}).
\end{aligned}$$

This form is useful because it shows that the predicted value depends on two estimated quantities: $\bar{Y}$ and $\widehat{\beta}_1$.

First consider uncertainty about the **mean response** at $X = x_0$, namely

$$\mathbb{E}[Y_0 \mid X_0 = x_0] = \beta_0 + \beta_1 x_0.$$

The variance of $\widehat{y}_0$ is

$$\begin{aligned}
\operatorname{Var}(\widehat{y}_0 \mid X_1, \dots, X_n) &= \operatorname{Var}\left(\bar{Y} + \widehat{\beta}_1(x_0 - \bar{X}) \;\middle|\; X_1, \dots, X_n\right) \\
&= \operatorname{Var}(\bar{Y} \mid X_1, \dots, X_n) + (x_0 - \bar{X})^2 \operatorname{Var}(\widehat{\beta}_1 \mid X_1, \dots, X_n) \\
&\quad + 2(x_0 - \bar{X}) \operatorname{Cov}(\bar{Y}, \widehat{\beta}_1 \mid X_1, \dots, X_n).
\end{aligned}$$

We have $\operatorname{Var}(\bar{Y} \mid X_1, \dots, X_n) = \sigma^2/n$ because $\bar{Y}$ averages n independent errors. From Equation 1.4, $\operatorname{Var}(\widehat{\beta}_1 \mid X_1, \dots, X_n) = \sigma^2/S_{xx}$. It remains to check the covariance term. Since

$$\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{\epsilon}, \qquad \widehat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X})\epsilon_i}{S_{xx}},$$

the only random parts are $\bar{\epsilon}$ and the weighted average of errors. Thus,

$$\begin{aligned}
\operatorname{Cov}(\bar{Y}, \widehat{\beta}_1 \mid X_1, \dots, X_n) &= \operatorname{Cov}\left(\bar{\epsilon}, \frac{\sum_{i=1}^n (X_i - \bar{X})\epsilon_i}{S_{xx}} \;\middle|\; X_1, \dots, X_n\right) \\
&= \frac{1}{nS_{xx}} \sum_{i=1}^n (X_i - \bar{X}) \operatorname{Var}(\epsilon_i \mid X_1, \dots, X_n) \\
&= \frac{\sigma^2}{nS_{xx}} \sum_{i=1}^n (X_i - \bar{X}) \\
&= 0.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\operatorname{Var}(\widehat{y}_0 \mid X_1, \dots, X_n) &= \frac{\sigma^2}{n} + (x_0 - \bar{X})^2 \frac{\sigma^2}{S_{xx}} \\
&= \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{X})^2}{S_{xx}}\right]. \qquad (1.5)
\end{aligned}$$

Replacing σ by $\widehat{\sigma}$ gives the standard error for the fitted mean response:

$$SE(\widehat{y}_0) = \widehat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{X})^2}{S_{xx}}}.$$

So a 95% confidence interval for the mean response is

$$\widehat{y}_0 \pm t_{n-2,0.975} \widehat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{X})^2}{S_{xx}}}.$$

This interval describes uncertainty about the average 2021 batting average among players with 2020 batting average x_0.

Now consider predicting one **individual** player with $X = x_0$. A new observation has the form

$$Y_0 = \beta_0 + \beta_1 x_0 + \epsilon_0,$$

where ϵ_0 is the new player's error term. The prediction error is

$$Y_0 - \widehat{y}_0 = (\beta_0 + \beta_1 x_0 + \epsilon_0) - \widehat{y}_0.$$

There are now two sources of randomness:

1. uncertainty in the fitted mean $\widehat{y}_0$;
2. the new player's individual error ϵ_0.

Because ϵ_0 is independent of the data used to fit the regression line,

$$\begin{aligned}
\mathrm{Var}(Y_0 - \widehat{y}_0 \mid X_1, \ldots, X_n,\ X_0 = x_0) &= \mathrm{Var}(\widehat{y}_0 \mid X_1, \ldots, X_n) + \mathrm{Var}(\epsilon_0) \\
&= \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{X})^2}{S_{xx}} \right] + \sigma^2 \\
&= \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{X})^2}{S_{xx}} \right]. \qquad (1.6)
\end{aligned}$$

Replacing σ by $\widehat{\sigma}$ gives a 95% prediction interval for an individual player:

$$\widehat{y}_0 \pm t_{n-2,0.975} \widehat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{X})^2}{S_{xx}}}.$$

The prediction interval is wider than the confidence interval for the mean response because of the extra 1 inside the square root. That extra term represents the individual player's own random deviation from the regression line. This distinction is crucial: predicting the **average** player with a given 2020 BA is easier than predicting one specific player.

In `R`, the same distinction appears in the `interval` argument:

```
# choose a new player with 2020 BA equal to .300
new_player = data.frame(BA_2020 = 0.300)

# uncertainty for the mean 2021 BA among .300 hitters
predict(model, newdata = new_player, interval = "confidence")

# uncertainty for one specific .300 hitter
predict(model, newdata = new_player, interval = "prediction")
```

Figure 1.4 summarizes the key uncertainty idea: the confidence band for the mean response is narrower than the prediction band for an individual player.

1.8 Remarks on Correlation and Model Checking

1.8.1 Correlation Can Be Misleading

Correlation is a measure of **linear association** between two variables. Figure 1.5 shows four plots with equal correlation, but they have **very** different relationships. What does this tell us?

Correlation can be **misleading** if:

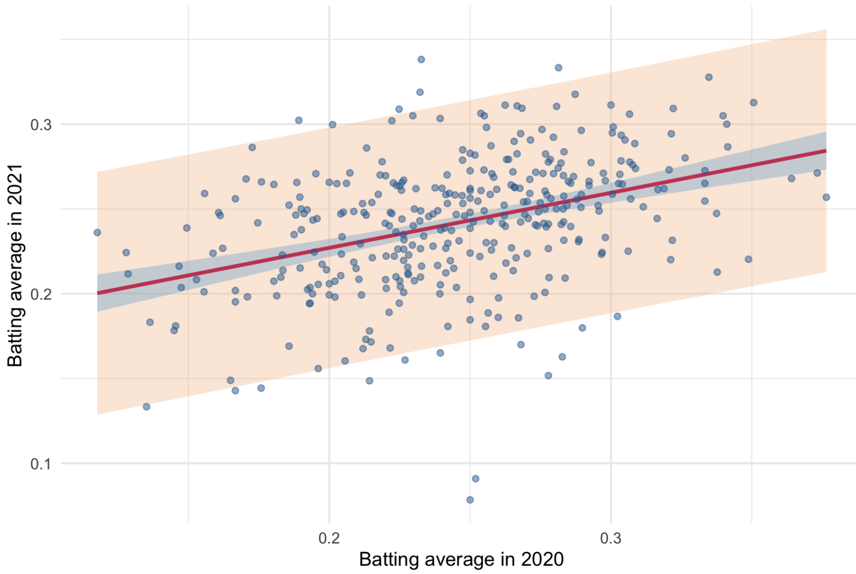

Figure 1.4: Red line is the fitted mean response; the inner blue band is the 95% confidence band for the mean response, and the wider orange band is the 95% prediction band for an individual player.

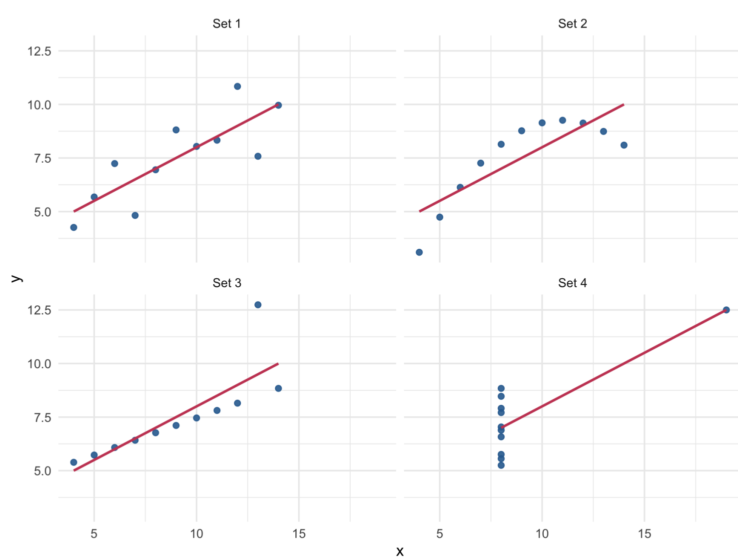

Figure 1.5: Each panel shows a different point pattern with nearly the same sample correlation; red lines are separate least-squares fits in each panel.

- the relationship is not linear;
- there are extreme outliers in the data;
- the data combine groups with different relationships.

It is best to use correlation to describe data whose scatter plots are roughly elliptical, or football-shaped. The lesson? **Always plot your data.**

1.8.2 Takeaways

At the end of this lecture, the simple linear regression workflow is:

1. **Start with a question.** Identify the response Y, the predictor X, and the prediction or explanation goal.
2. **Plot the data.** Use the scatterplot to look for direction, strength, linearity, outliers, and possible subgroup structure.
3. **Specify the model.** Write a model for the conditional mean,
$$\mathbb{E}[Y_i \mid X_i] = \beta_0 + \beta_1 X_i,$$
together with assumptions about the error terms.
4. **Fit the line.** Use ordinary least squares to estimate β_0 and β_1 by minimizing the residual sum of squares.
5. **Interpret the estimates.** Explain the slope in the units of the problem and connect it to correlation through
$$\widehat{\beta}_1 = r_{XY}\frac{S_Y}{S_X}.$$
6. **Standardize when useful.** Standardizing X changes the slope to a "per one-standard-deviation increase in X" interpretation; standardizing both X and Y makes the slope equal to the sample correlation.
7. **Report uncertainty.** Distinguish uncertainty about the fitted mean response from uncertainty about an individual prediction.
8. **Check the model.** Use plots and residuals to look for nonlinearity, outliers, changing variance, and hidden groups.

Appendix A Motivation via Normal Random Variables

This appendix gives a probability-based motivation for the regression formula. The main lecture did not require this material, but it explains why the slope takes the form "correlation times a ratio of standard deviations" in a common population model.

We begin by motivating linear regression through the decomposition of jointly distributed random variables. We begin with the case of jointly standard normal variables.

A.1 Jointly Standard Normal Random Variables

Definition 1.8 (Jointly Standard Normal Variables)**.** *Let (X, Y) be jointly normal random variables with*

$$\mathbb{E}[X] = \mathbb{E}[Y] = 0, \quad \mathrm{Var}(X) = \mathrm{Var}(Y) = 1, \quad \mathrm{Cov}(X, Y) = \rho.$$

Then (X, Y) are jointly standard normal.

Theorem 1.9 (Decomposition of Jointly Standard Normal Variables)**.** *Assume $|\rho| < 1$. For jointly standard normal variables (X, Y) as defined above, we can write*

$$Y = \rho X + \sqrt{1-\rho^2}Z,$$

where $Z \sim \mathcal{N}(0, 1)$ and $X \perp\!\!\!\perp Z$ (X is independent of Z).

Proof. Define Z as follows:

$$Z = \frac{Y - \rho X}{\sqrt{1-\rho^2}}. \tag{1.7}$$

We can verify that:

- $\mathbb{E}[Z] = 0$:

$$\begin{aligned}\mathbb{E}[Z] &= \frac{\mathbb{E}[Y] - \rho\mathbb{E}[X]}{\sqrt{1-\rho^2}} \\ &= \frac{0 - \rho \cdot 0}{\sqrt{1-\rho^2}} = 0.\end{aligned}$$

- $\mathrm{Var}(Z) = 1$:

$$\begin{aligned}\mathrm{Var}(Z) &= \frac{\mathrm{Var}(Y - \rho X)}{1-\rho^2} \\ &= \frac{\mathrm{Var}(Y) + \rho^2\,\mathrm{Var}(X) - 2\rho\,\mathrm{Cov}(X, Y)}{1-\rho^2} \\ &= \frac{1 + \rho^2 - 2\rho^2}{1-\rho^2} = 1.\end{aligned}$$

- $\mathrm{Cov}(X, Z) = 0$:

$$\begin{aligned}\mathrm{Cov}(X, Z) &= \frac{\mathrm{Cov}(X, Y) - \rho\,\mathrm{Var}(X)}{\sqrt{1-\rho^2}} \\ &= \frac{\rho - \rho}{\sqrt{1-\rho^2}} = 0.\end{aligned}$$

Since Z and X are jointly normal and uncorrelated, they are independent. The result follows by solving for Y:

$$Y = \rho X + \sqrt{1-\rho^2}Z.$$

$\square$

Theorem 1.10 (Best Linear Predictor)**.** *For jointly standard normal variables, the best linear predictor, or least-squares predictor, of Y given X is*

$$\widehat{Y} = \rho X.$$

Proof. We aim to minimize the mean squared error:

$$\min_a \mathbb{E}\left[(Y - aX)^2\right].$$

Expanding, we have

$$\begin{aligned}
\mathbb{E}\left[(Y - aX)^2\right] &= \mathbb{E}\left[Y^2 - 2aXY + a^2X^2\right] \\
&= \mathbb{E}[Y^2] - 2a\mathbb{E}[XY] + a^2\mathbb{E}[X^2] \\
&= 1 - 2a\rho + a^2.
\end{aligned}$$

Setting the derivative to zero,

$$\begin{aligned}
\frac{\partial}{\partial a}\mathbb{E}\left[(Y - aX)^2\right] &= -2\rho + 2a = 0 \\
\implies a &= \rho.
\end{aligned}$$

So the best linear predictor is

$$\widehat{Y} = \rho X.$$

$\square$

In general, the best linear predictor and the conditional expectation need not be the same. In the jointly normal case, they are the same.

Theorem 1.11 (Conditional Expectation)**.** *For jointly standard normal variables, the conditional expectation $\mathbb{E}[Y \mid X]$ equals the least-squares predictor:*

$$\mathbb{E}[Y \mid X] = \rho X.$$

Proof. From the decomposition $Y = \rho X + \sqrt{1-\rho^2}Z$ where $X \perp\!\!\!\perp Z$,

$$\mathbb{E}[Y \mid X] = \rho X + \sqrt{1-\rho^2}\mathbb{E}[Z \mid X].$$

By independence of X and Z,

$$\mathbb{E}[Y \mid X] = \rho X + \sqrt{1-\rho^2}\mathbb{E}[Z].$$

Since $Z \sim \mathcal{N}(0,1)$,

$$\begin{aligned}
\mathbb{E}[Y \mid X] &= \rho X + \sqrt{1-\rho^2} \cdot 0 \\
&= \rho X.
\end{aligned}$$

$\square$

We will now generalize this result to non-standard normal variables.

A.2 Generalization to Non-Standard Normals

Suppose now that
$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2), \quad \mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y.$$
Letting $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$, define standardized versions of X and Y as
$$X^* = \frac{X - \mu_X}{\sigma_X}, \quad Y^* = \frac{Y - \mu_Y}{\sigma_Y}.$$
Then (X^*, Y^*) are jointly standard normal with correlation ρ, and from Theorem 1.9,
$$Y^* = \rho X^* + \sqrt{1-\rho^2}Z, \quad Z \sim \mathcal{N}(0,1), \ X^* \perp\!\!\!\perp Z.$$
Multiplying both sides by σ_Y and adding μ_Y, we obtain
$$Y = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(X - \mu_X) + \sigma_Y\sqrt{1-\rho^2}Z.$$
And then similar to Theorem 1.11,
$$\mathbb{E}[Y \mid X] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(X - \mu_X),$$
and the error term is independent Gaussian noise.

A.3 A Linear Conditional Mean with Non-Normal X

Suppose X is a standardized random variable with $\mathbb{E}[X] = 0$ and $\mathrm{Var}(X) = 1$, but X is not necessarily Gaussian. Let
$$Y = \rho X + \sqrt{1-\rho^2}Z,$$
where $Z \sim \mathcal{N}(0,1)$ and $X \perp\!\!\!\perp Z$.

To compute the least-squares predictor of Y given X, we will need the following theorem.

Theorem 1.12 (No-Intercept Best Linear Predictor)**.** *If X is not constant and we restrict attention to predictors of the form aX, then*
$$\arg\min_a \mathbb{E}[(Y - aX)^2] = \frac{\mathbb{E}[XY]}{\mathbb{E}[X^2]}.$$
If $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, this equals
$$\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}.$$

The proof of this theorem is left as an exercise.

Then we compute the covariance of X and Y as follows:
$$\begin{aligned}
\mathrm{Cov}(X,Y) &= \mathrm{Cov}(X, \rho X + \sqrt{1-\rho^2}Z) \\
&= \mathrm{Cov}(X, \rho X) + \mathrm{Cov}(X, \sqrt{1-\rho^2}Z) \\
&= \rho\,\mathrm{Var}(X) + \sqrt{1-\rho^2}\,\mathrm{Cov}(X, Z) \\
&= \rho(1) + \sqrt{1-\rho^2}\cdot 0 \\
&= \rho.
\end{aligned}$$

Since $\mathrm{Var}(X) = 1$, the optimal coefficient is

$$a = \rho.$$

Thus, the best linear predictor of Y given X is still

$$\widehat{Y} = \rho X.$$

Furthermore, $\mathbb{E}[Y \mid X] = \rho X$ still holds because

$$\begin{aligned}
\mathbb{E}[Y \mid X] &= \mathbb{E}[\rho X + \sqrt{1-\rho^2} Z \mid X] \\
&= \rho X + \sqrt{1-\rho^2}\,\mathbb{E}[Z \mid X] \\
&= \rho X + \sqrt{1-\rho^2}\,\mathbb{E}[Z] \\
&= \rho X.
\end{aligned}$$

So even if X is non-Gaussian, the regression relationship remains the same due to the model structure.

References

[Lahman] Friendly, M., Dalzell, C., Monkman, M., Murphy, et al., Lahman: Sean 'Lahman' Baseball Database, 2024. R package version 12.0-0.